

TYPES OF FACTOR MODELS

Factor models can be divided into four categories of models: index models, macroeconomic models, cross-sectional or fundamental data models, and statistical factor models. These are described below.

Table 6.6 provides a comparison of the common risk factor models used in the industry. A description of each is provided below.

Table 6.6 Risk Factor Models				
Volatility Model	Types	Formula	Parameter	Calculation
Time Series	Index, Macro, Technical	$R = \hat{B}F + \varepsilon$	$\hat{B}$	OLS
Cross-Sectional	Fundamental	$R = B\hat{F} + \varepsilon$	$\hat{F}$	OLS
Statistical/PCA	Implicit Factors	$R = \hat{B}\hat{F} + \varepsilon$	$\hat{B}, \hat{F}$	OLS, MLE
	Data Driven	$\bar{C}$		Eigenvalue, SVD Factor Analysis

Index Model

There are two forms of the index model commonly used in the industry: single index and multi-index models. The single index model is based on a single major market index such as the SP500. The same index is used as the input factor across all stocks. The multi-index model commonly incorporates the general market index, the stock's sector index, and additionally, the stock's industry index. The market index will be the same for all stocks but the sector index and industry index will be different based on the company's economic grouping. All stocks in the same sector will use the same sector index, and all stocks in the same industry will use the same industry index.

Single Index Model

The simplest of all the multi-factor models is the single index model. This model formulates a relationship between stock returns and market movement. In most situations, the SP500 index or some other broad market index is used as a proxy for the whole market.